HAT-P-36b

R-0680

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-36_260412` · created 2026-07-27T19:26:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461142.7014 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1283 +/- 0.0097
Transit depth (Rp/Rs)^2	1.64 +/- 0.25 [%]
Orbital Inclination (inc)	89.43 +/- 2.02
Airmass coefficient 1 (a1)	1.203 +/- 0.012
Airmass coefficient 2 (a2)	-0.177 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	None
Optimal Aperture	3.67
Optimal Annulus	9.2
Transit Duration (day)	0.0972 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1283 ± 0.0097 vs 0.126 (deviation 0.24σ)
Duration fit vs published	0.0972 d vs 0.0925 d (deviation 2.94σ)
Mid-time O–C	11.7 min
Depth SNR	13.2
Residual scatter	1.03 %

Light curve

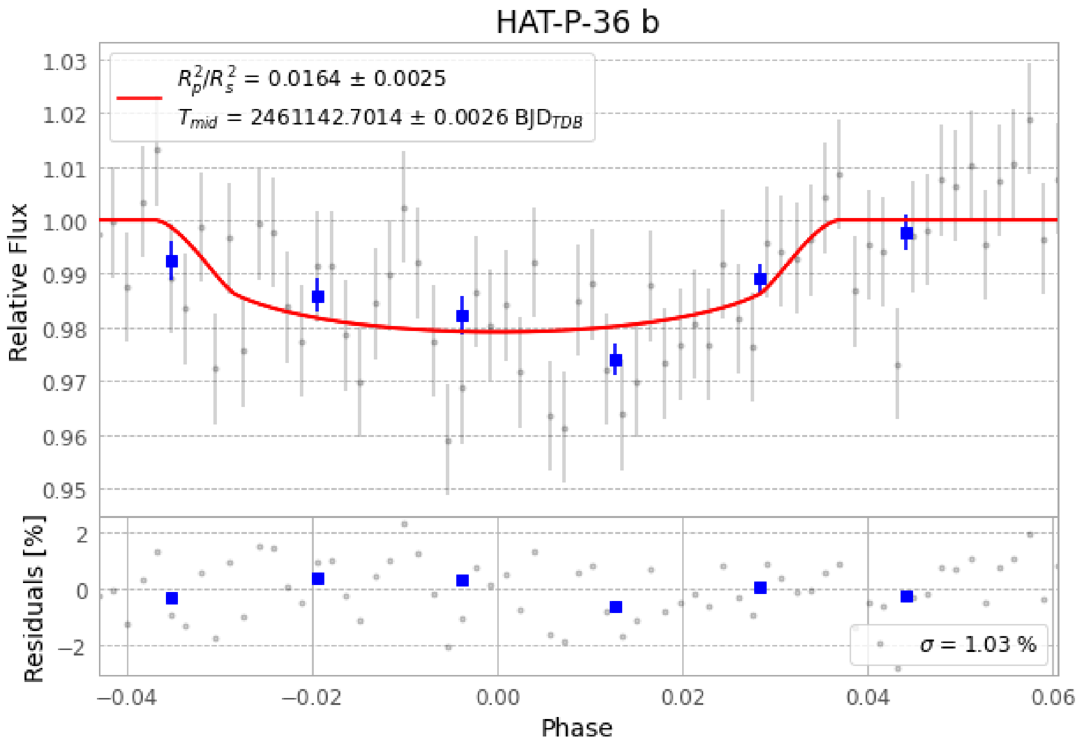

Figure 1 — FinalLightCurve_HAT-P-36 b_2026-04-11.png (SHA-256 44ec5f8094d4ace5…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 227], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c09e01335dfdbcc9…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813`

Data provenance

67 FITS frames (44 MB), `HATP-36260412032115.FITS` → `HATP-36260412063915.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260412.log (a5fe2c864369…), FinalLightCurve_HAT-P-36 b_2026-04-11.png (44ec5f8094d4…), FinalLightCurve_HAT-P-36 b_2026-04-11.csv (41c9acdf2a70…)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 19:26Z.